

# Value of Information Analysis

## METIMMOX-1 Economic Evaluation

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### Table of contents

|           |                                                              |           |
|-----------|--------------------------------------------------------------|-----------|
| <b>1</b>  | <b>Overview</b>                                              | <b>2</b>  |
| 1.1       | Population-Level Context . . . . .                           | 2         |
| 1.2       | Treatment-Biomarker Interaction Parameters . . . . .         | 2         |
| <b>2</b>  | <b>Model Structure Definitions</b>                           | <b>4</b>  |
| <b>3</b>  | <b>Scenario Definitions</b>                                  | <b>5</b>  |
| <b>4</b>  | <b>Base Case Scenario</b>                                    | <b>6</b>  |
| 4.1       | EVPI Comparison Across Model Structures . . . . .            | 6         |
| 4.2       | EVPPPI Results by Model Structure . . . . .                  | 8         |
| 4.3       | Interpretation . . . . .                                     | 8         |
| 4.3.1     | Population-Level Research Value . . . . .                    | 8         |
| <b>5</b>  | <b>Scenario 1: Decreased Nivolumab Cost</b>                  | <b>9</b>  |
| 5.1       | EVPI Comparison Across Model Structures . . . . .            | 9         |
| 5.2       | EVPPPI Results by Model Structure . . . . .                  | 9         |
| <b>6</b>  | <b>Scenario 2: Higher WTP Threshold (EUR100,000)</b>         | <b>11</b> |
| 6.1       | EVPI Comparison Across Model Structures . . . . .            | 11        |
| 6.2       | EVPPPI Results by Model Structure . . . . .                  | 11        |
| <b>7</b>  | <b>Scenario 3: Higher WTP with Decreased Nivolumab Cost</b>  | <b>13</b> |
| 7.1       | EVPI Comparison Across Model Structures . . . . .            | 13        |
| 7.2       | EVPPPI Results by Model Structure . . . . .                  | 13        |
| <b>8</b>  | <b>Scenario 4: Highest WTP Threshold (EUR150,000)</b>        | <b>15</b> |
| 8.1       | EVPI Comparison Across Model Structures . . . . .            | 15        |
| 8.2       | EVPPPI Results by Model Structure . . . . .                  | 15        |
| <b>9</b>  | <b>Scenario 5: Highest WTP with Decreased Nivolumab Cost</b> | <b>17</b> |
| 9.1       | EVPI Comparison Across Model Structures . . . . .            | 17        |
| 9.2       | EVPPPI Results by Model Structure . . . . .                  | 17        |
| <b>10</b> | <b>Comparative Analysis Across Scenarios and Models</b>      | <b>19</b> |
| 10.1      | EVPI Comparison . . . . .                                    | 19        |
| 10.2      | EVPPPI Parameter Heatmap by Model Structure . . . . .        | 20        |
| 10.3      | Summary Comparison Table . . . . .                           | 21        |

|                                                                |    |
|----------------------------------------------------------------|----|
| 10.4 Population-Level EVPPI Across Scenarios . . . . .         | 21 |
| 10.5 Key Findings . . . . .                                    | 21 |
| 10.5.1 Model Structure Robustness . . . . .                    | 21 |
| 10.5.2 Scenario with Greatest Decision Uncertainty . . . . .   | 22 |
| 10.5.3 Most Important Parameter Across All Scenarios . . . . . | 22 |
| 10.5.4 General Observations . . . . .                          | 22 |

# 1 Overview

This report presents the results of an Expected Value of Partially Perfect Information (EVPPI) analysis for the METIMMOX-1 decision-analytic model. The analysis evaluates the value of reducing uncertainty in individual model parameters across multiple scenarios with varying willingness-to-pay (WTP) thresholds and nivolumab drug costs.

**Key Feature:** This report compares EVPI and EVPPI results across three different model structures (Model A, B, C) to assess the robustness of value of information findings to modeling assumptions about how biomarker interactions are specified.

The Expected Value of Partially Perfect Information (EVPPI) quantifies the value of eliminating uncertainty in specific model parameters. It represents the maximum amount that should be paid for additional research to perfectly inform a particular parameter. EVPI (Expected Value of Perfect Information) represents the value of eliminating all uncertainty, and EVPPI for any parameter cannot exceed total EVPI.

## 1.1 Population-Level Context

To inform research prioritization decisions, EVPPI values are scaled from per-patient estimates to the Norwegian eligible population:

- **Annual incidence:** 1500 MSS/pMMR mCRC patients per year
- **Research horizon:** 10 years
- **Discount rate:** 3.5% for future research benefits
- **Effective population size:** 12475 discounted patient-years

Population-level EVPPI represents the total expected value (in millions of EUR) of eliminating uncertainty about a parameter or parameter group across the entire eligible population over the research horizon. This scaling accounts for the discounted flow of future patients who would benefit from the research.

**Data sources:** The annual incidence estimate is based on Norwegian Cancer Registry data (approximately 4,000 annual CRC cases) and international literature estimates for MSS/pMMR proportion (70-75% of metastatic cases) and metastatic presentation rates (approximately 50%), yielding a conservative estimate of 1,500 eligible patients per year.

## 1.2 Treatment-Biomarker Interaction Parameters

In addition to cost, utility, and prevalence parameters, this analysis evaluates the EVPPI for treatment-biomarker interaction coefficients extracted from the survival models. These parameters (prefixed **b\_**) represent the effect modification of experimental treatment by biomarker status – the core clinical question underlying biomarker-guided therapy.

The interaction coefficients are extracted from the same resampled survival models used in the PSA, ensuring alignment between coefficient variation and outcome variation. For each biomarker strategy, the interaction coefficient is extracted from that strategy’s own resampled

model (e.g., `b_crp_rx_os` from `sampling_models$crp`). Both OS and PFS interaction coefficients are analyzed individually and as per-biomarker groups.

High EVPPI for interaction parameters would indicate that better estimates of differential treatment benefit by biomarker status would significantly reduce decision uncertainty – directly informing future trial design and biomarker validation studies.

## 2 Model Structure Definitions

We compare three model structures to assess the robustness of value of information results to the specification of biomarker-treatment interactions in the survival models.

Table 1: Model structure definitions used in the analysis

| Model   | Name     | Formula Pattern                                        | Description                                                                                  |
|---------|----------|--------------------------------------------------------|----------------------------------------------------------------------------------------------|
| Model A | Joint    | Age + sex + Rx + crp*Rx + tlr*Rx + tmb_braf*Rx         | All biomarkers and all treatment interactions in one unified model                           |
| Model B | Focused  | Age + sex + Rx + crp + tmb_braf + crp:Rx + tmb_braf:Rx | Per-strategy formulas:<br>CRP/TMB_BRAF with both interactions, TLR with TLR-only interaction |
| Model C | Separate | Age + sex + Rx + [biomarker]:Rx                        | Only one biomarker and its treatment interaction per model (simplest specification)          |

**Rationale:** Model A (Joint) is the most comprehensive specification but may suffer from over-fitting with limited sample size. Model C (Separate) is the simplest but ignores potential confounding from other biomarkers. Model B (Focused) uses a shared formula with CRP and TMB/BRAF (excluding TLR), providing simultaneous estimates for both interactions while adjusting for each other. Comparing results across all three structures provides insight into the structural uncertainty of our findings.

### 3 Scenario Definitions

Table 2: Analysis scenario definitions. Higher WTP thresholds (EUR100,000 and EUR150,000) are arbitrary exploratory thresholds, though the highest severity adjustment in Norwegian guidelines may fall between these values. The decreased nivolumab cost (EUR4,641) represents an 80% discount compared to the 40% discount in the base case scenario.

| Scenario Name                      | WTP Threshold | Nivolumab Cost |
|------------------------------------|---------------|----------------|
| Base Case                          | EUR 51,000    | EUR13,923      |
| Decreased Nivolumab Cost           | EUR 51,000    | EUR 4,641      |
| WTP €100,000                       | EUR100,000    | EUR13,923      |
| WTP €100,000 + Decreased Nivolumab | EUR100,000    | EUR 4,641      |
| WTP €150,000                       | EUR150,000    | EUR13,923      |
| WTP €150,000 + Decreased Nivolumab | EUR150,000    | EUR 4,641      |

## 4 Base Case Scenario

**Willingness-to-Pay Threshold:** EUR51,000

**Nivolumab Cost:** EUR13,923

### 4.1 EVPI Comparison Across Model Structures

Table 3: Expected Value of Perfect Information (EVPI) by model structure - Base Case

|    | Model             | EVPI       |
|----|-------------------|------------|
| 5  | Model A: Joint    | EUR1079.15 |
| 56 | Model B: Focused  | EUR1112.9  |
| 57 | Model C: Separate | EUR713.53  |

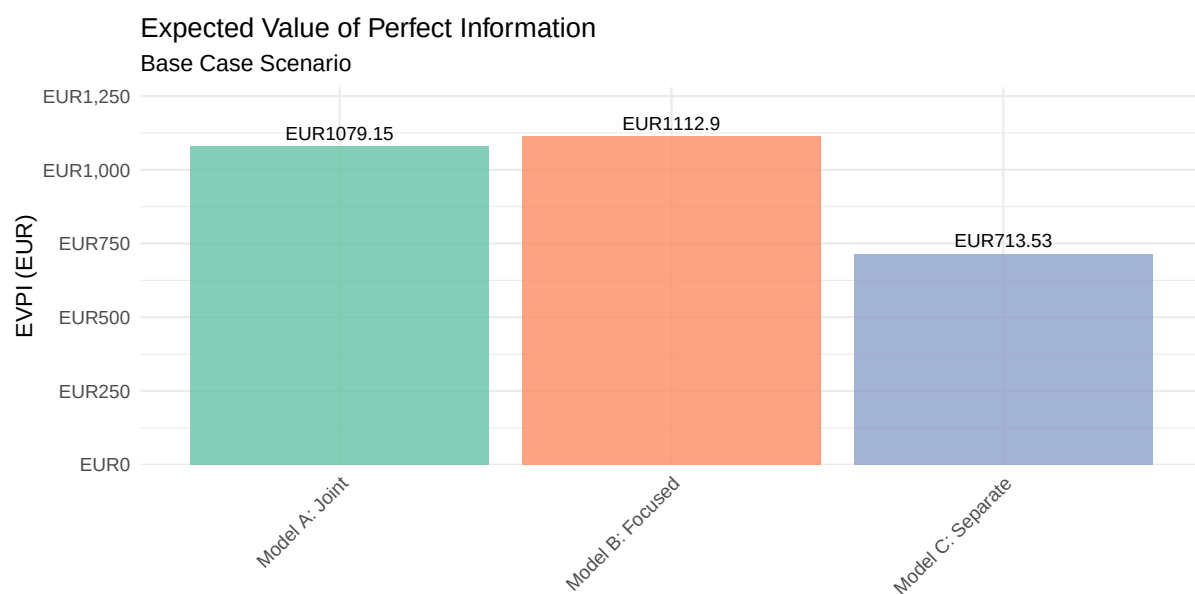


Figure 1: EVPI comparison across model structures (Base Case)

Table 4: Per-patient EVPPI results by model structure - Base Case (ranked by average importance)

| Parameter        | EVPPI (A)  | % (A) | EVPPI (B) | % (B) | EVPPI (C) | % (C) |
|------------------|------------|-------|-----------|-------|-----------|-------|
| u_np             | EUR1079.15 | 100%  | EUR988.53 | 88.8% | EUR713.53 | 100%  |
| u_p              | EUR951.69  | 88.2% | EUR942.07 | 84.6% | EUR713.53 | 100%  |
| c_drug_FLOX      | EUR219.46  | 20.3% | EUR161.25 | 14.5% | EUR8.1    | 1.1%  |
| c_other_last     | EUR96.04   | 8.9%  | EUR102.45 | 9.2%  | EUR77.66  | 10.9% |
| c_drug_nivo      | EUR66.2    | 6.1%  | EUR16     | 1.4%  | EUR31.6   | 4.4%  |
| c_test_blood     | EUR64.02   | 5.9%  | EUR14.65  | 1.3%  | EUR19.96  | 2.8%  |
| c_test_CT        | EUR25.63   | 2.4%  | EUR4.32   | 0.4%  | EUR46.37  | 6.5%  |
| c_other_baseline | EUR61.73   | 5.7%  | EUR0      | 0%    | EUR0      | 0%    |
| c_test_NGS       | EUR0       | 0%    | EUR60.62  | 5.4%  | EUR0      | 0%    |
| c_other_visit    | EUR0       | 0%    | EUR0      | 0%    | EUR0      | 0%    |
| c_other_follow   | EUR0       | 0%    | EUR0      | 0%    | EUR0      | 0%    |

Table 5: Population-level EVPPI results by model structure - Base Case (millions of EUR over 10 years, discounted at 3.5%, ranked by average importance)

| Parameter        | Pop. EVPPI (A) | % (A) | Pop. EVPPI (B) | % (B) | Pop. EVPPI (C) | % (C) |
|------------------|----------------|-------|----------------|-------|----------------|-------|
| u_np             | EUR13.46M      | 100%  | EUR12.33M      | 88.8% | EUR8.90M       | 100%  |
| u_p              | EUR11.87M      | 88.2% | EUR11.75M      | 84.6% | EUR8.90M       | 100%  |
| c_drug_FLOX      | EUR2.74M       | 20.3% | EUR2.01M       | 14.5% | EUR0.10M       | 1.1%  |
| c_other_last     | EUR1.20M       | 8.9%  | EUR1.28M       | 9.2%  | EUR0.97M       | 10.9% |
| c_drug_nivo      | EUR0.83M       | 6.1%  | EUR0.20M       | 1.4%  | EUR0.39M       | 4.4%  |
| c_test_blood     | EUR0.80M       | 5.9%  | EUR0.18M       | 1.3%  | EUR0.25M       | 2.8%  |
| c_test_CT        | EUR0.32M       | 2.4%  | EUR0.05M       | 0.4%  | EUR0.58M       | 6.5%  |
| c_other_baseline | EUR0.77M       | 5.7%  | EUR0.00M       | 0%    | EUR0.00M       | 0%    |
| c_test_NGS       | EUR0.00M       | 0%    | EUR0.76M       | 5.4%  | EUR0.00M       | 0%    |
| c_other_visit    | EUR0.00M       | 0%    | EUR0.00M       | 0%    | EUR0.00M       | 0%    |
| c_other_follow   | EUR0.00M       | 0%    | EUR0.00M       | 0%    | EUR0.00M       | 0%    |

## 4.2 EVPPI Results by Model Structure

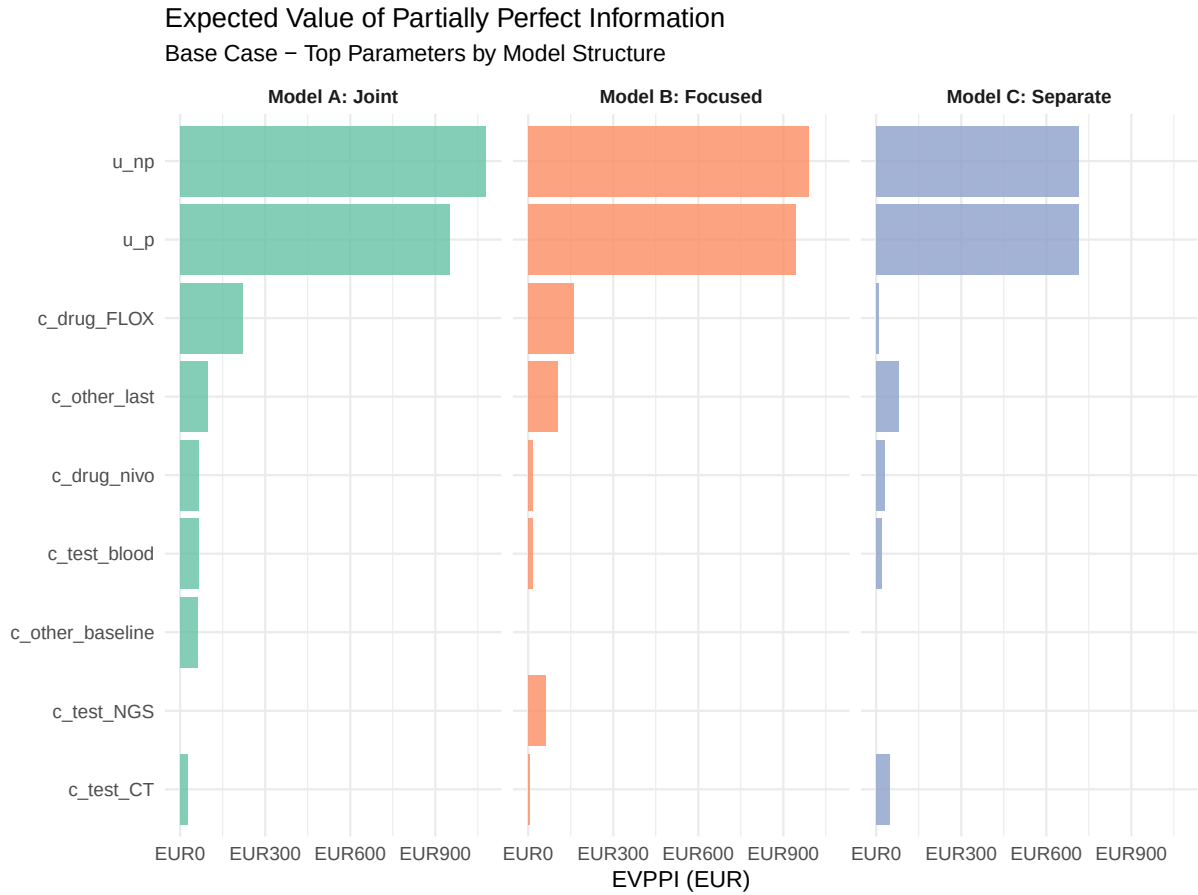


Figure 2: EVPPI by parameter and model structure (Base Case)

## 4.3 Interpretation

### 4.3.1 Population-Level Research Value

**Per-Patient Level:** Utility parameters (u\_np, u\_p) show the highest per-patient EVPPI, ranging from approximately EUR800-1,100 in the base case, while cost parameters show lower EVPPI (typically <EUR200), reflecting less uncertainty in cost estimates.

**Population Level:** Scaling to the Norwegian eligible population (1500 patients/year over 10 years, discounted at 3.5%), utility parameter uncertainty translates to **EUR13.5 million** in potential research value. This population-level value provides context for research investment decisions. For example, a study to reduce uncertainty in utility values (e.g., a large-scale quality-of-life measurement study in this patient population) could be justified if its cost is substantially less than EUR13.5 million.

Cost parameters, while showing lower per-patient EVPPI, still represent substantial population-level values (EUR0.8 million for nivolumab cost), particularly relevant for market access negotiations and price negotiations.

The positive EVPI indicates decision uncertainty at the base case WTP threshold. This suggests that additional research to reduce parameter uncertainty could have value. The comparison across model structures shows whether this finding is robust to the specification of biomarker-treatment interactions.

## 5 Scenario 1: Decreased Nivolumab Cost

**Willingness-to-Pay Threshold:** EUR51,000

**Nivolumab Cost:** EUR4,641 (80% discount)

### 5.1 EVPI Comparison Across Model Structures

Table 6: Expected Value of Perfect Information (EVPI) by model structure - Scenario 1

|     | Model             | EVPI       |
|-----|-------------------|------------|
| 53  | Model A: Joint    | EUR7201.02 |
| 531 | Model B: Focused  | EUR7303.79 |
| 532 | Model C: Separate | EUR7260.81 |

### 5.2 EVPPI Results by Model Structure

Table 7: EVPPI results by model structure - Scenario 1: Decreased Nivolumab Cost

| Parameter        | EVPPI (A) | % (A) | EVPPI (B) | % (B) | EVPPI (C) | % (C) |
|------------------|-----------|-------|-----------|-------|-----------|-------|
| u_np             | EUR882.33 | 12.3% | EUR931.31 | 12.8% | EUR891.48 | 12.3% |
| u_p              | EUR677.34 | 9.4%  | EUR845.63 | 11.6% | EUR843.7  | 11.6% |
| c_other_last     | EUR102.42 | 1.4%  | EUR67.5   | 0.9%  | EUR103.76 | 1.4%  |
| c_drug_nivo      | EUR0      | 0%    | EUR110.91 | 1.5%  | EUR0      | 0%    |
| c_other_follow   | EUR0      | 0%    | EUR94.44  | 1.3%  | EUR0      | 0%    |
| c_test_NGS       | EUR53.35  | 0.7%  | EUR0      | 0%    | EUR11.87  | 0.2%  |
| c_other_visit    | EUR0      | 0%    | EUR0      | 0%    | EUR58.21  | 0.8%  |
| c_test_CT        | EUR38.78  | 0.5%  | EUR0      | 0%    | EUR0      | 0%    |
| c_test_blood     | EUR0      | 0%    | EUR0      | 0%    | EUR27.41  | 0.4%  |
| c_drug_FLOX      | EUR0      | 0%    | EUR0      | 0%    | EUR23.98  | 0.3%  |
| c_other_baseline | EUR20.11  | 0.3%  | EUR0      | 0%    | EUR0      | 0%    |

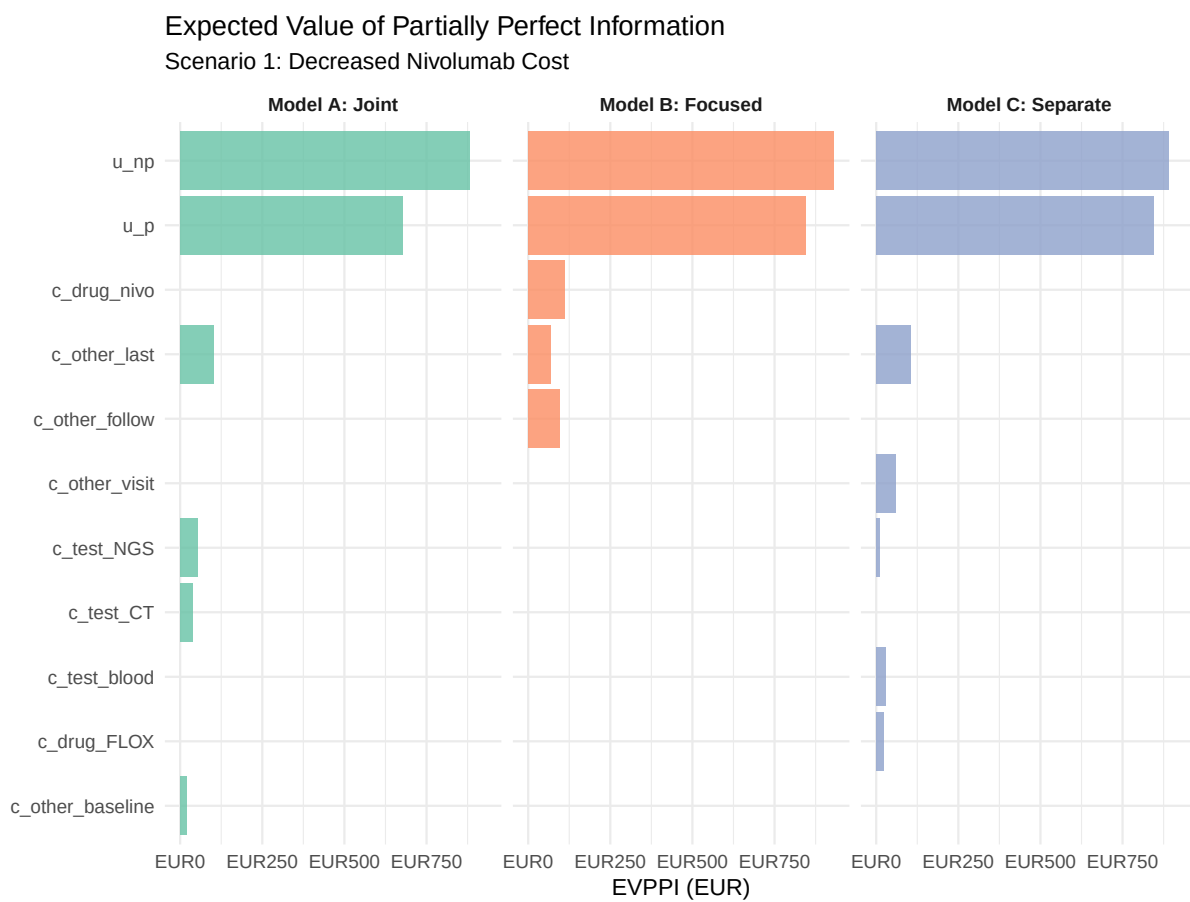


Figure 3: EVPPPI by parameter and model structure (Scenario 1)

## 6 Scenario 2: Higher WTP Threshold (EUR100,000)

**Willingness-to-Pay Threshold:** EUR1e+05

**Nivolumab Cost:** EUR13,923 (standard)

### 6.1 EVPI Comparison Across Model Structures

Table 8: Expected Value of Perfect Information (EVPI) by model structure - Scenario 2

|     | Model             | EVPI       |
|-----|-------------------|------------|
| 51  | Model A: Joint    | EUR8809.37 |
| 511 | Model B: Focused  | EUR9287.77 |
| 612 | Model C: Separate | EUR8410.69 |

### 6.2 EVPPI Results by Model Structure

Table 9: EVPPI results by model structure - Scenario 2: WTP EUR100,000

| Parameter        | EVPPI (A)  | % (A) | EVPPI (B)  | % (B) | EVPPI (C)  | % (C) |
|------------------|------------|-------|------------|-------|------------|-------|
| u_np             | EUR2155.28 | 24.5% | EUR1936.37 | 20.8% | EUR1823.41 | 21.7% |
| u_p              | EUR1871.05 | 21.2% | EUR1834.81 | 19.8% | EUR1877.06 | 22.3% |
| c_drug_FLOX      | EUR382.87  | 4.3%  | EUR275.49  | 3%    | EUR0       | 0%    |
| c_other_last     | EUR127.79  | 1.5%  | EUR140.15  | 1.5%  | EUR93.23   | 1.1%  |
| c_drug_nivo      | EUR148.08  | 1.7%  | EUR27.39   | 0.3%  | EUR58.63   | 0.7%  |
| c_test_blood     | EUR130.83  | 1.5%  | EUR14.06   | 0.2%  | EUR25.15   | 0.3%  |
| c_test_CT        | EUR35.51   | 0.4%  | EUR2.23    | 0%    | EUR78.5    | 0.9%  |
| c_other_baseline | EUR110.49  | 1.3%  | EUR0       | 0%    | EUR0       | 0%    |
| c_test_NGS       | EUR0       | 0%    | EUR103.55  | 1.1%  | EUR0       | 0%    |
| c_other_visit    | EUR0       | 0%    | EUR0       | 0%    | EUR0       | 0%    |
| c_other_follow   | EUR0       | 0%    | EUR0       | 0%    | EUR0       | 0%    |

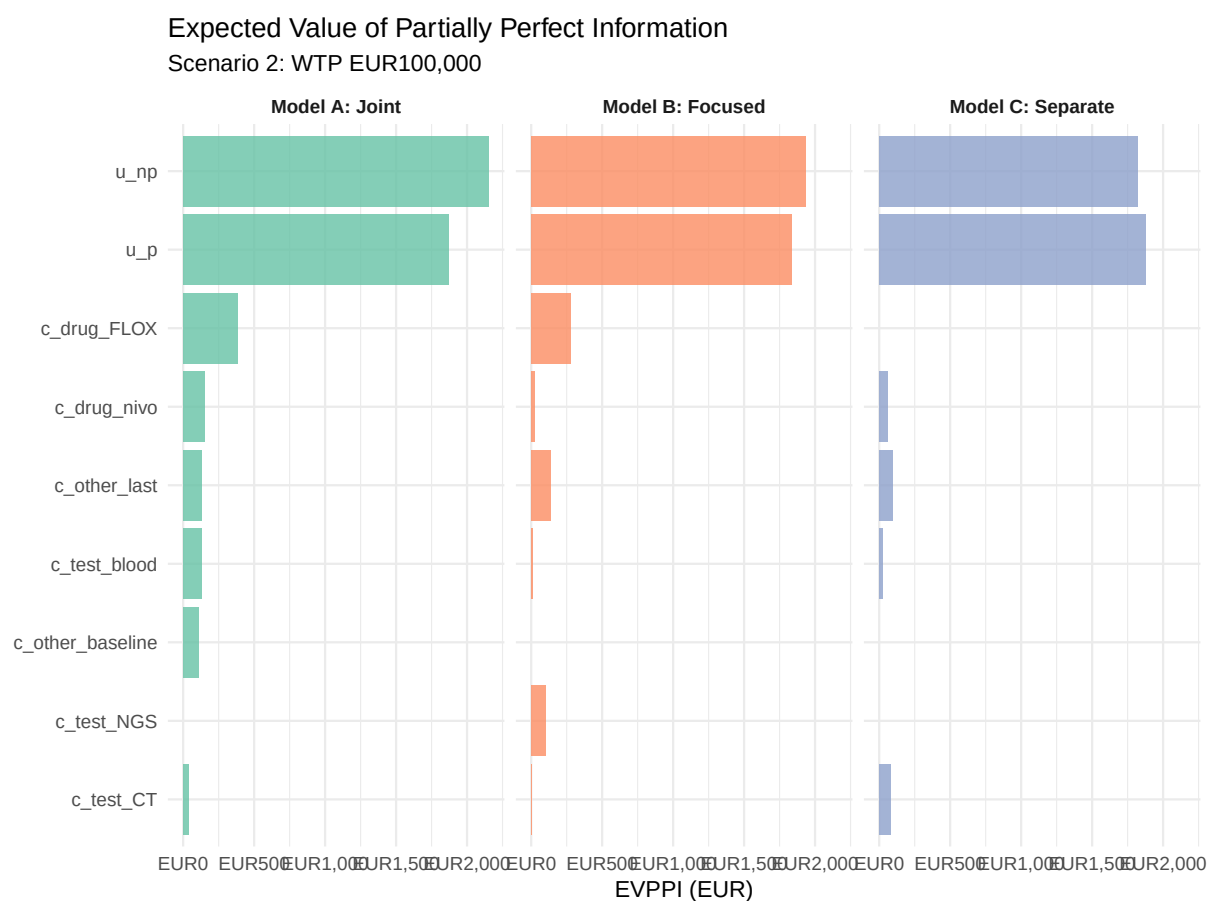


Figure 4: EVVPI by parameter and model structure (Scenario 2)

## 7 Scenario 3: Higher WTP with Decreased Nivolumab Cost

**Willingness-to-Pay Threshold:** EUR1e+05

**Nivolumab Cost:** EUR4,641 (80% discount)

### 7.1 EVPI Comparison Across Model Structures

Table 10: Expected Value of Perfect Information (EVPI) by model structure - Scenario 3

|     | Model             | EVPI        |
|-----|-------------------|-------------|
| 54  | Model A: Joint    | EUR21762.79 |
| 541 | Model B: Focused  | EUR21614.53 |
| 542 | Model C: Separate | EUR21762.49 |

### 7.2 EVPPI Results by Model Structure

Table 11: EVPPI results by model structure - Scenario 3: WTP EUR100,000 + Decreased Nivolumab

| Parameter        | EVPPI (A)  | % (A) | EVPPI (B)  | % (B) | EVPPI (C)  | % (C) |
|------------------|------------|-------|------------|-------|------------|-------|
| u_np             | EUR2643.74 | 12.1% | EUR3373.99 | 15.6% | EUR3207.95 | 14.7% |
| u_p              | EUR2020.86 | 9.3%  | EUR2260.7  | 10.5% | EUR1766.22 | 8.1%  |
| c_drug_nivo      | EUR710.59  | 3.3%  | EUR932.07  | 4.3%  | EUR695.48  | 3.2%  |
| c_other_last     | EUR538.64  | 2.5%  | EUR561.74  | 2.6%  | EUR233.13  | 1.1%  |
| c_other_visit    | EUR614.05  | 2.8%  | EUR376.58  | 1.7%  | EUR141.53  | 0.7%  |
| c_drug_FLOX      | EUR486.12  | 2.2%  | EUR477.12  | 2.2%  | EUR120.28  | 0.6%  |
| c_other_follow   | EUR471.09  | 2.2%  | EUR488.54  | 2.3%  | EUR0       | 0%    |
| c_test_NGS       | EUR575.12  | 2.6%  | EUR283.86  | 1.3%  | EUR99.97   | 0.5%  |
| c_other_baseline | EUR628.08  | 2.9%  | EUR311.24  | 1.4%  | EUR0       | 0%    |
| c_test_CT        | EUR501.17  | 2.3%  | EUR244.4   | 1.1%  | EUR0       | 0%    |
| c_test_blood     | EUR161.56  | 0.7%  | EUR448.85  | 2.1%  | EUR11.7    | 0.1%  |

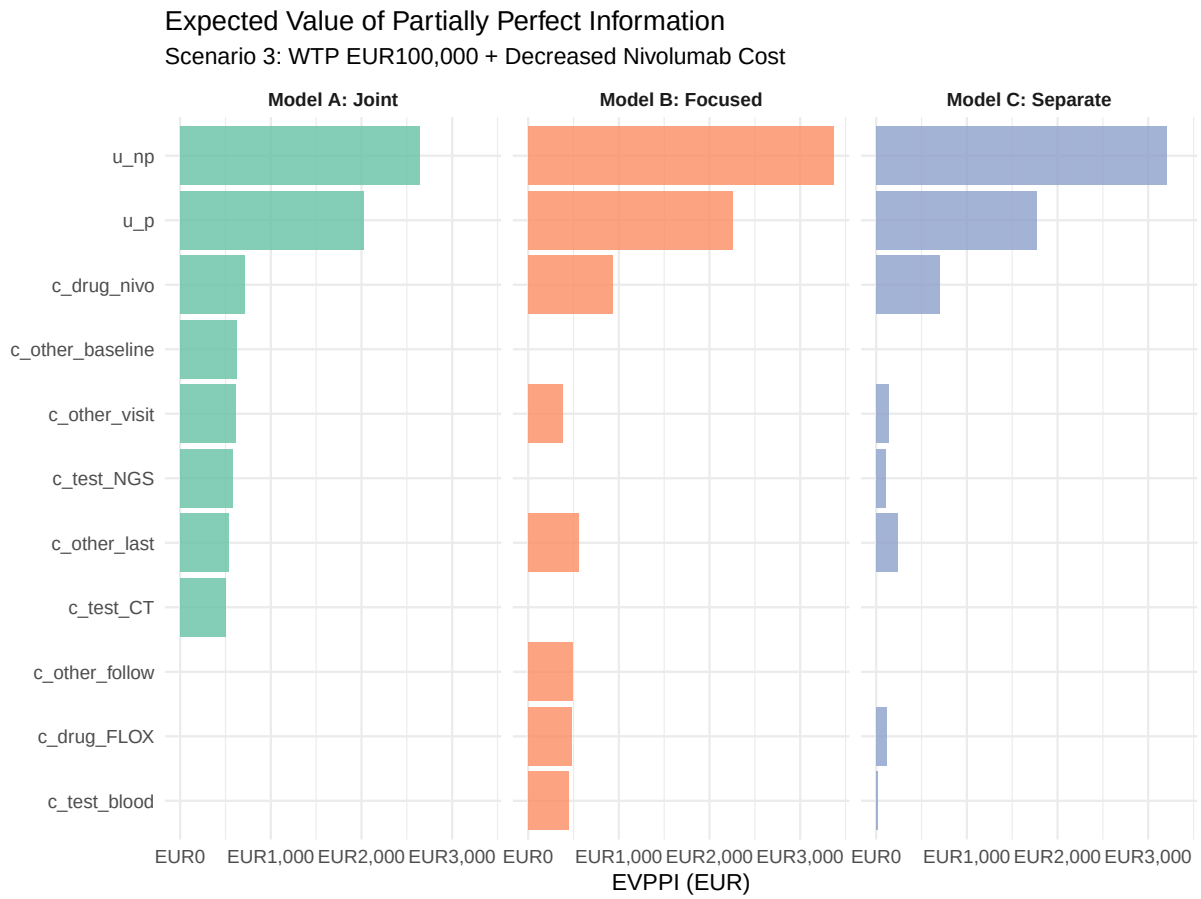


Figure 5: EVPPPI by parameter and model structure (Scenario 3)

## 8 Scenario 4: Highest WTP Threshold (EUR150,000)

**Willingness-to-Pay Threshold:** EUR150,000

**Nivolumab Cost:** EUR13,923 (standard)

### 8.1 EVPI Comparison Across Model Structures

Table 12: Expected Value of Perfect Information (EVPI) by model structure - Scenario 4

|     | Model             | EVPI        |
|-----|-------------------|-------------|
| 52  | Model A: Joint    | EUR21320.55 |
| 521 | Model B: Focused  | EUR21793.4  |
| 622 | Model C: Separate | EUR21660.6  |

### 8.2 EVPPI Results by Model Structure

Table 13: EVPPI results by model structure - Scenario 4: WTP EUR150,000

| Parameter        | EVPPI (A)  | % (A) | EVPPI (B)  | % (B) | EVPPI (C)  | % (C) |
|------------------|------------|-------|------------|-------|------------|-------|
| u_np             | EUR3235.46 | 15.2% | EUR2903.55 | 13.3% | EUR2738.96 | 12.6% |
| u_p              | EUR2809.18 | 13.2% | EUR2745.77 | 12.6% | EUR2815.2  | 13%   |
| c_drug_FLOX      | EUR549.61  | 2.6%  | EUR392.07  | 1.8%  | EUR0       | 0%    |
| c_other_last     | EUR160.18  | 0.8%  | EUR178.62  | 0.8%  | EUR109.13  | 0.5%  |
| c_drug_nivo      | EUR231.62  | 1.1%  | EUR39.01   | 0.2%  | EUR86.22   | 0.4%  |
| c_test_blood     | EUR199.01  | 0.9%  | EUR13.46   | 0.1%  | EUR30.45   | 0.1%  |
| c_other_baseline | EUR160.24  | 0.8%  | EUR0       | 0%    | EUR0       | 0%    |
| c_test_CT        | EUR45.6    | 0.2%  | EUR0.1     | 0%    | EUR111.28  | 0.5%  |
| c_test_NGS       | EUR0       | 0%    | EUR147.36  | 0.7%  | EUR0       | 0%    |
| c_other_visit    | EUR0       | 0%    | EUR0       | 0%    | EUR0       | 0%    |
| c_other_follow   | EUR0       | 0%    | EUR0       | 0%    | EUR0       | 0%    |

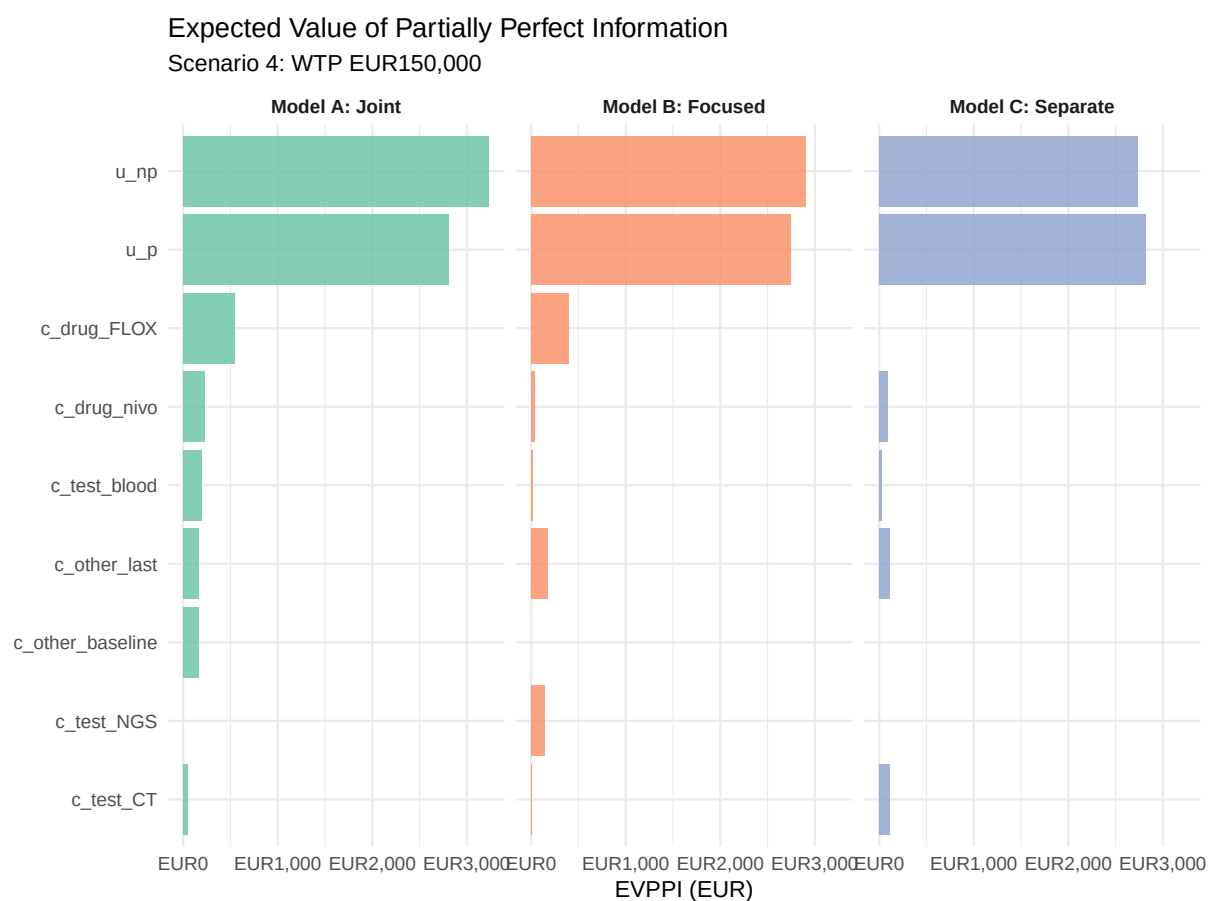


Figure 6: EVPPPI by parameter and model structure (Scenario 4)

## 9 Scenario 5: Highest WTP with Decreased Nivolumab Cost

**Willingness-to-Pay Threshold:** EUR150,000

**Nivolumab Cost:** EUR4,641 (80% discount)

### 9.1 EVPI Comparison Across Model Structures

Table 14: Expected Value of Perfect Information (EVPI) by model structure - Scenario 5

|     | Model             | EVPI        |
|-----|-------------------|-------------|
| 55  | Model A: Joint    | EUR30046.55 |
| 551 | Model B: Focused  | EUR31487.26 |
| 552 | Model C: Separate | EUR29940.71 |

### 9.2 EVPPI Results by Model Structure

Table 15: EVPPI results by model structure - Scenario 5: WTP EUR150,000 + Decreased Nivolumab

| Parameter        | EVPPI (A)  | % (A) | EVPPI (B)  | % (B) | EVPPI (C)  | % (C) |
|------------------|------------|-------|------------|-------|------------|-------|
| u_np             | EUR3082.79 | 10.3% | EUR3962.38 | 12.6% | EUR3908.8  | 13.1% |
| u_p              | EUR2403.67 | 8%    | EUR1988.96 | 6.3%  | EUR2015.32 | 6.7%  |
| c_other_last     | EUR285.79  | 1%    | EUR0       | 0%    | EUR248.3   | 0.8%  |
| c_other_visit    | EUR0       | 0%    | EUR66.13   | 0.2%  | EUR197.08  | 0.7%  |
| c_test_CT        | EUR213     | 0.7%  | EUR34.47   | 0.1%  | EUR0       | 0%    |
| c_test_NGS       | EUR86.23   | 0.3%  | EUR0       | 0%    | EUR132.07  | 0.4%  |
| c_drug_nivo      | EUR0       | 0%    | EUR21.05   | 0.1%  | EUR121.87  | 0.4%  |
| c_other_follow   | EUR133.83  | 0.4%  | EUR0       | 0%    | EUR0       | 0%    |
| c_other_baseline | EUR72.34   | 0.2%  | EUR0       | 0%    | EUR0       | 0%    |
| c_drug_FLOX      | EUR0       | 0%    | EUR0       | 0%    | EUR63.37   | 0.2%  |
| c_test_blood     | EUR0       | 0%    | EUR0       | 0%    | EUR0       | 0%    |

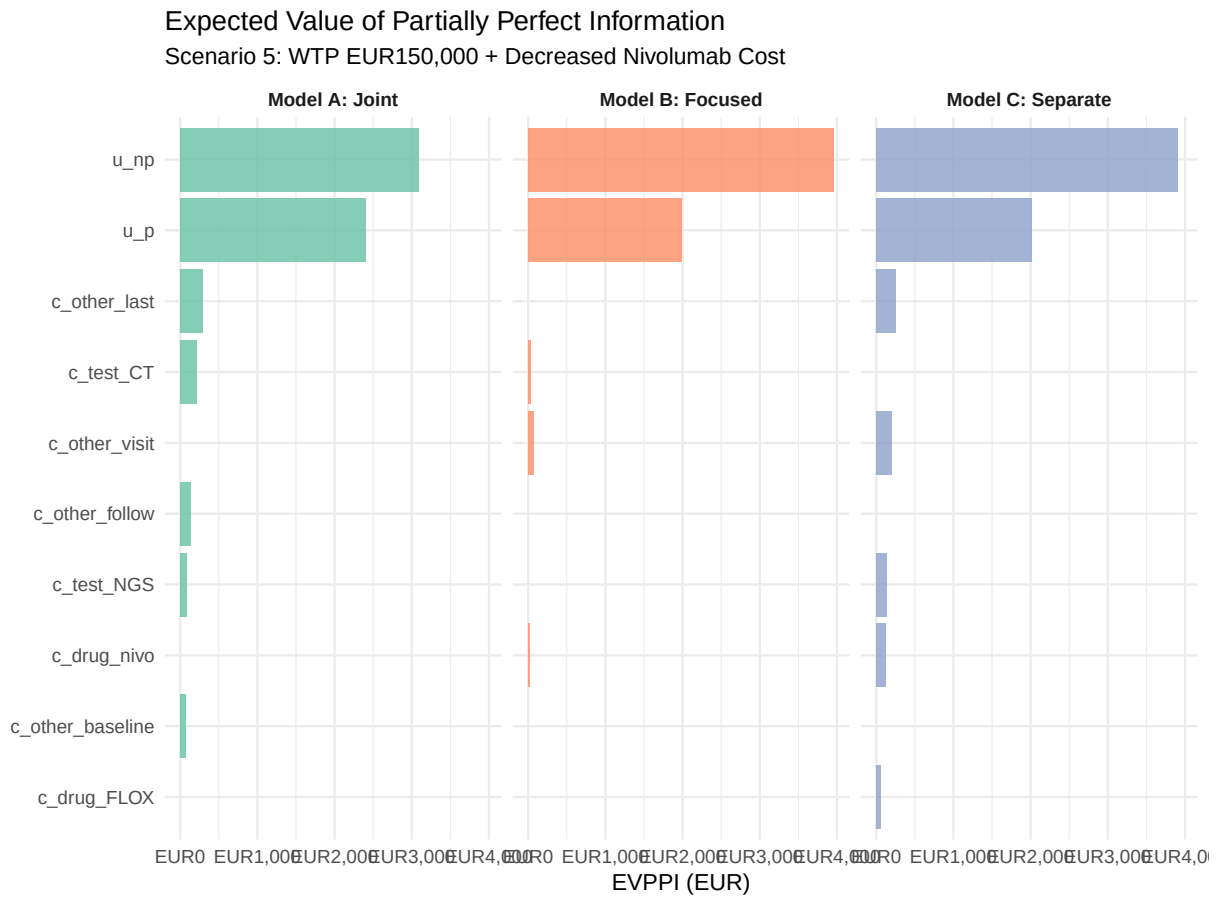


Figure 7: EVVPI by parameter and model structure (Scenario 5)

## 10 Comparative Analysis Across Scenarios and Models

### 10.1 EVPI Comparison

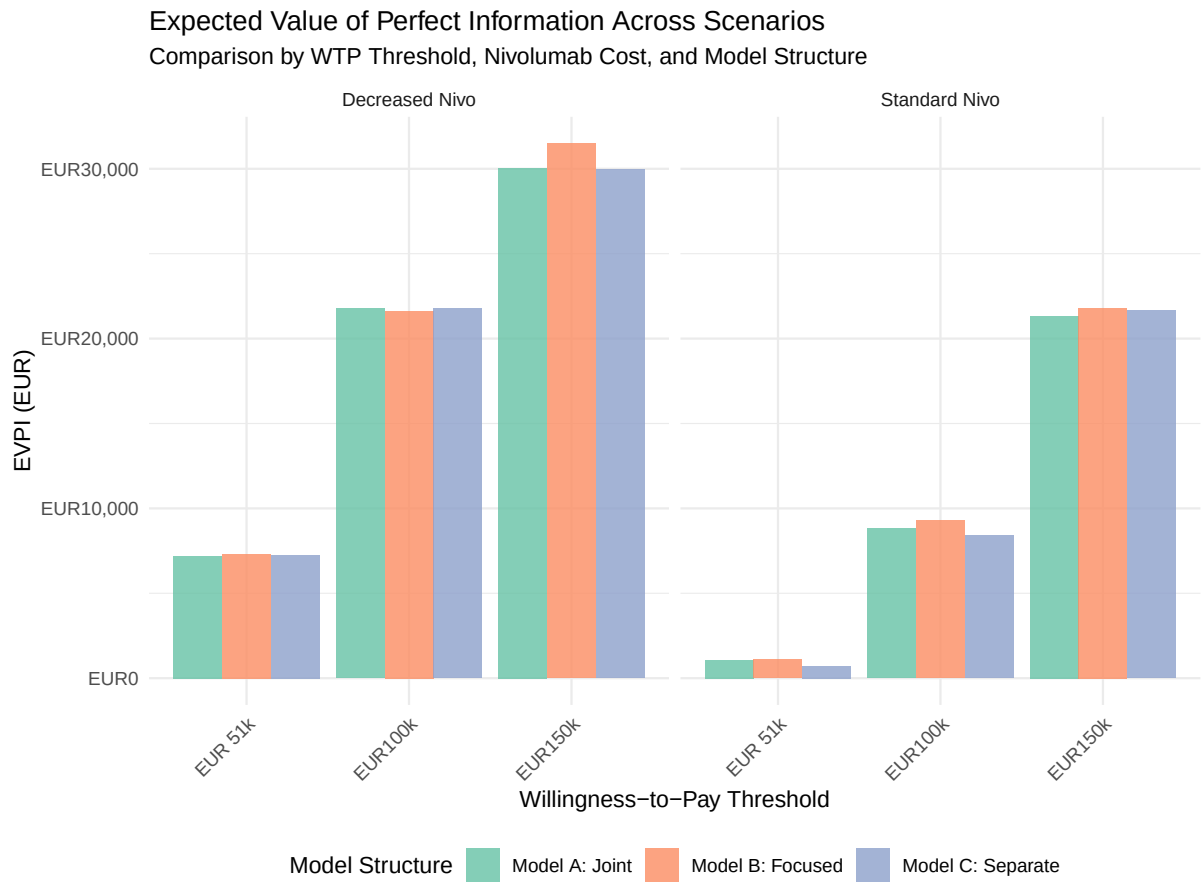


Figure 8: Expected Value of Perfect Information across scenarios and model structures

## 10.2 EVPPI Parameter Heatmap by Model Structure

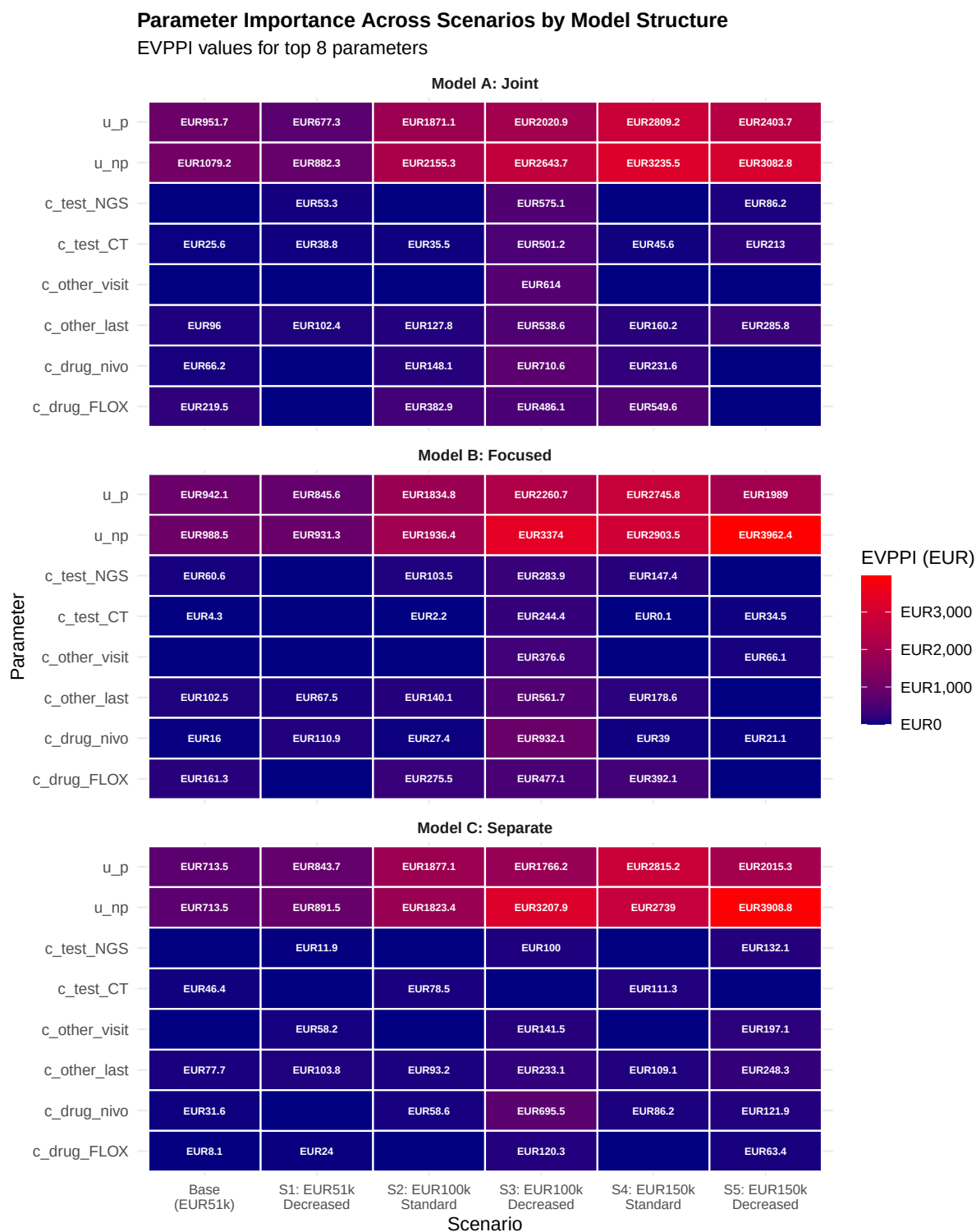


Figure 9: Parameter importance across scenarios by model structure (top 8 parameters)

Table 16: Summary of EVPI and top EVPPI by model structure and scenario

| Model             | Scenario                           | WTP        | EVPI        | Top Parameter | Max EVPPI  | % of EVPI |
|-------------------|------------------------------------|------------|-------------|---------------|------------|-----------|
| Model A: Joint    | Base Case                          | EUR 51,000 | EUR1079.15  | u_np          | EUR1079.15 | 100.0     |
| Model A: Joint    | Decreased Nivolumab Cost           | EUR 51,000 | EUR7201.02  | u_np          | EUR882.33  | 12.3      |
| Model A: Joint    | WTP €100,000                       | EUR100,000 | EUR8809.37  | u_np          | EUR2155.28 | 24.5      |
| Model A: Joint    | WTP €100,000 + Decreased Nivolumab | EUR100,000 | EUR21762.79 | u_np          | EUR2643.74 | 12.1      |
| Model A: Joint    | WTP €150,000                       | EUR150,000 | EUR21320.55 | u_np          | EUR3235.46 | 15.2      |
| Model A: Joint    | WTP €150,000 + Decreased Nivolumab | EUR150,000 | EUR30046.55 | u_np          | EUR3082.79 | 10.3      |
| Model B: Focused  | Base Case                          | EUR 51,000 | EUR1112.9   | u_np          | EUR988.53  | 88.8      |
| Model B: Focused  | Decreased Nivolumab Cost           | EUR 51,000 | EUR7303.79  | u_np          | EUR931.31  | 12.8      |
| Model B: Focused  | WTP €100,000                       | EUR100,000 | EUR9287.77  | u_np          | EUR1936.37 | 20.8      |
| Model B: Focused  | WTP €100,000 + Decreased Nivolumab | EUR100,000 | EUR21614.53 | u_np          | EUR3373.99 | 15.6      |
| Model B: Focused  | WTP €150,000                       | EUR150,000 | EUR21793.4  | u_np          | EUR2903.55 | 13.3      |
| Model B: Focused  | WTP €150,000 + Decreased Nivolumab | EUR150,000 | EUR31487.26 | u_np          | EUR3962.38 | 12.6      |
| Model C: Separate | Base Case                          | EUR 51,000 | EUR713.53   | u_np          | EUR713.53  | 100.0     |
| Model C: Separate | Decreased Nivolumab Cost           | EUR 51,000 | EUR7260.81  | u_np          | EUR891.48  | 12.3      |
| Model C: Separate | WTP €100,000                       | EUR100,000 | EUR8410.69  | u_p           | EUR1877.06 | 22.3      |
| Model C: Separate | WTP €100,000 + Decreased Nivolumab | EUR100,000 | EUR21762.49 | u_np          | EUR3207.95 | 14.7      |
| Model C: Separate | WTP €150,000                       | EUR150,000 | EUR21660.6  | u_p           | EUR2815.2  | 13.0      |
| Model C: Separate | WTP €150,000 + Decreased Nivolumab | EUR150,000 | EUR29940.71 | u_np          | EUR3908.8  | 13.1      |

Table 17: Population-level EVPPI (EUR millions) for key parameters across scenarios and model structures. Based on 1500 patients/year over 10 years, discounted at 3.5%.

| Scenario                           | Parameter   | Model A: Joint | Model B: Focused | Model C: Separate |
|------------------------------------|-------------|----------------|------------------|-------------------|
| Base Case                          | u_np        | 13.46          | 12.33            | 8.90              |
| Base Case                          | u_p         | 11.87          | 11.75            | 8.90              |
| Base Case                          | c_drug_nivo | 0.83           | 0.20             | 0.39              |
| Decreased Nivolumab Cost           | u_np        | 11.01          | 11.62            | 11.12             |
| Decreased Nivolumab Cost           | u_p         | 8.45           | 10.55            | 10.53             |
| Decreased Nivolumab Cost           | c_drug_nivo | 0.00           | 1.38             | 0.00              |
| WTP €100,000                       | u_np        | 26.89          | 24.16            | 22.75             |
| WTP €100,000                       | u_p         | 23.34          | 22.89            | 23.42             |
| WTP €100,000                       | c_drug_nivo | 1.85           | 0.34             | 0.73              |
| WTP €100,000 + Decreased Nivolumab | u_np        | 32.98          | 42.09            | 40.02             |
| WTP €100,000 + Decreased Nivolumab | u_p         | 25.21          | 28.20            | 22.03             |
| WTP €100,000 + Decreased Nivolumab | c_drug_nivo | 8.86           | 11.63            | 8.68              |
| WTP €150,000                       | u_np        | 40.36          | 36.22            | 34.17             |
| WTP €150,000                       | u_p         | 35.04          | 34.25            | 35.12             |
| WTP €150,000                       | c_drug_nivo | 2.89           | 0.49             | 1.08              |
| WTP €150,000 + Decreased Nivolumab | u_np        | 38.46          | 49.43            | 48.76             |
| WTP €150,000 + Decreased Nivolumab | u_p         | 29.99          | 24.81            | 25.14             |
| WTP €150,000 + Decreased Nivolumab | c_drug_nivo | 0.00           | 0.26             | 1.52              |

### 10.3 Summary Comparison Table

### 10.4 Population-Level EVPPI Across Scenarios

### 10.5 Key Findings

#### 10.5.1 Model Structure Robustness

**Parameter importance is consistent across model structures.** The parameter ‘u\_np’ is the most important driver of value of information across all three model specifications. This suggests that conclusions about research priorities are robust to the choice of model structure.

### 10.5.2 Scenario with Greatest Decision Uncertainty

- **Model A: Joint:** WTP €150,000 + Decreased Nivolumab (EVPI = EUR30046.55)
- **Model B: Focused:** WTP €150,000 + Decreased Nivolumab (EVPI = EUR31487.26)
- **Model C: Separate:** WTP €150,000 + Decreased Nivolumab (EVPI = EUR29940.71)

### 10.5.3 Most Important Parameter Across All Scenarios

**u\_np** (Total EVPPI = EUR40459.01)

### 10.5.4 General Observations

**Impact of WTP Threshold:** Higher WTP thresholds generally increase decision uncertainty (EVPI), creating greater value for additional research to reduce parameter uncertainty.

**Impact of Nivolumab Cost:** Decreased nivolumab costs shift the cost-effectiveness landscape and may change which parameters drive decision uncertainty.

**Impact of Model Structure:** Comparing EVPI and EVPPI across model structures helps identify whether value of information conclusions are robust to structural uncertainty in the survival model specification.

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**Note on population-level EVPPI:** Population-level values calculated assuming 1500 annual eligible MSS/pMMR mCRC patients in Norway (based on Cancer Registry data and international prevalence estimates), discounted at 3.5% per year over a 10-year research horizon.